Derivatives: Contracts, Models, and Markets · Technical Note 3

The Minimum-Variance Hedge Ratio

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Section 3.5 of Derivatives: Contracts, Models, and Markets states $h^* = \rho\,\sigma_S/\sigma_F$ in a box and puts it to work without proving it. This note derives it, records the variance it achieves, extends it to several futures contracts at once, and obtains the tailing adjustment from daily settlement rather than asserting it. It closes with the question the boxed formula hides: what the ratio costs when its three inputs are estimated instead of known. That cost decides something the formula alone cannot—whether a second hedging instrument is worth using at all—and for the chapter's own example the answer is no.

1 THE PROBLEM

A hedger holds an exposure that will be worth ΔS per unit more or less than expected, and takes a futures position of h units per unit of exposure. Per unit of exposure the hedged outcome changes by

$$\Delta\Pi = \Delta S - h\,\Delta F \tag{TN3.1}$$

with $h > 0$ for a short hedge—long the asset, short futures—and the signs reversed for a long hedge. Nothing below depends on which.

Write $\sigma_S^2 = \mathrm{Var}(\Delta S)$, $\sigma_F^2 = \mathrm{Var}(\Delta F)$ and ρ for their correlation, all measured over the life of the hedge. Then

$$v(h) \;=\; \mathrm{Var}(\Delta\Pi) \;=\; \sigma_S^2 - 2h\rho\,\sigma_S\sigma_F + h^2\sigma_F^2 \tag{TN3.2}$$

This is a quadratic in h with a positive leading coefficient, so it has one stationary point and that point is its minimum. Differentiating,

$$v'(h) = -2\rho\,\sigma_S\sigma_F + 2h\sigma_F^2 = 0 \qquad \Longrightarrow \qquad \boxed{h^* = \rho\,\frac{\sigma_S}{\sigma_F}}$$

and $v''(h) = 2\sigma_F^2 > 0$ confirms it. That is (3.3). *No distributional assumption was used*—not normality, not independence over time, not even that the moments come from a model. Only that the two variances are finite and $\sigma_F > 0$.

Substituting h^* back into (TN3.2) gives the variance the hedge attains:

$$v(h^*) = \sigma_S^2\left(1 - \rho^2\right) \tag{TN3.3}$$

So the fraction of variance the hedge removes is ρ^2, and nothing the hedger does can improve on it with this one contract. The imperfection is a property of the pair of assets, not of the hedger's skill. With $\rho = 0.8$, as in Example 3.3, the best attainable outcome leaves 36% of the variance in place.

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Two corollaries are worth stating because they are often assumed rather than checked. *A hedge always helps*, in the sense that $v(h^*) \le \sigma_S^2$ with equality only when $\rho = 0$. And *a one-for-one hedge can be worse than none at all*: $v(1) > \sigma_S^2$ whenever $\sigma_F > 2\rho\,\sigma_S$, which is exactly the case a cross hedger meets when the futures asset is much the more volatile of the two.

2 WHY IT IS A REGRESSION SLOPE

The ordinary least squares slope from regressing ΔS on ΔF is

$$\beta = \frac{\mathrm{Cov}(\Delta S, \Delta F)}{\mathrm{Var}(\Delta F)} = \frac{\rho\,\sigma_S\sigma_F}{\sigma_F^2} = \rho\,\frac{\sigma_S}{\sigma_F} = h^*$$

The two problems are the same problem. Least squares minimizes the variance of the residual $\Delta S - \beta\,\Delta F$; the hedger minimizes the variance of $\Delta S - h\,\Delta F$. They are the same objective written twice. The regression's residual variance is (TN3.3) and its R^2 is ρ^2, so *hedge effectiveness and R^2 are the same number*, which is why the chapter can quote one for the other.

Why the regression must be run on changes. Spot and futures price *levels* are both close to random walks. Regressing one on the other returns R^2 near one and a slope near one whether or not the two assets have anything to do with each other, because the fitted relationship is the shared trend rather than the co-movement a hedge exploits. The identity above is a statement about ΔS and ΔF; run on levels it is not the hedge ratio, and it flatters the hedge badly. This is Section 3.5's warning, and (TN3.2) is why it matters: the quantity being minimized is the variance of a *change*.

3 SEVERAL CONTRACTS AT ONCE

Suppose k futures contracts are available. Let $\Delta\mathbf{F}$ be the vector of their price changes, $\mathbf{\Sigma}$ its covariance matrix, and $\mathbf{c}$ the vector of covariances between ΔS and each contract. With a vector of ratios $\mathbf{h}$,

$$v(\mathbf{h}) = \sigma_S^2 - 2\,\mathbf{h}^\top\mathbf{c} + \mathbf{h}^\top\mathbf{\Sigma}\,\mathbf{h}$$

Setting the gradient to zero gives $-2\mathbf{c} + 2\mathbf{\Sigma h} = 0$, so

$$\mathbf{h}^* = \mathbf{\Sigma}^{-1}\mathbf{c}, \qquad v(\mathbf{h}^*) = \sigma_S^2 - \mathbf{c}^\top\mathbf{\Sigma}^{-1}\mathbf{c} \tag{TN3.4}$$

which reduces to h^* and (TN3.3) when $k = 1$. It is the multiple regression of ΔS on all k contracts, for the same reason as before.

Example 3.1 A second contract, and what it is worth

Return to Example 3.3's airline: jet fuel with $\sigma_S = 0.032$, heating oil futures with $\sigma_1 = 0.040$ and $\rho_{S1} = 0.80$. Add crude oil futures with $\sigma_2 = 0.035$ and $\rho_{S2} = 0.75$. The answer turns entirely on ρ_{12}, how alike the two contracts are.

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ρ_{12}	h_1	h_2	**residual variance**	R^2
— (heating oil alone)	0.6400	—	3.6864×10^{-4}	0.6400
0.90	0.5263	0.1444	3.6379×10^{-4}	0.6447
0.70	0.4314	0.3406	2.9616×10^{-4}	0.7108
0.50	0.4533	0.4267	2.0139×10^{-4}	0.8033

Crude oil is highly correlated with jet fuel and yet adds almost nothing. At $\rho_{12} = 0.90$—which is realistic for two refined-product curves—the second contract lifts R^2 from 0.6400 to 0.6447, four-tenths of a percentage point, while splitting the position across two contracts and two basis risks.

What a second instrument contributes is not its correlation with the exposure but its independence of the first. At $\rho_{12} = 0.50$ the same crude contract, with the same $\rho_{S2} = 0.75$, takes R^2 to 0.8033. The hedged variance falls by nearly half, and h_2 rises from 0.14 to 0.43. Nothing about the second contract's relationship to jet fuel changed between those two rows.

4 WHAT ESTIMATION COSTS

Everything so far treats σ_S, σ_F and ρ as known. They are estimated, and the estimate is used as though it were the truth. The cost is computable.

Let $\hat{h}$ be the OLS slope from n observations, and write $\sigma_\varepsilon^2 = \sigma_S^2(1 - \rho^2)$ for the true residual variance. Conditional on the realized $\hat{h}$, the hedged variance is $v(\hat{h})$, and expanding (TN3.2) about h^*,

$$v(\hat{h}) = v(h^*) + \sigma_F^2 \, (\hat{h} - h^*)^2$$

because the linear term vanishes at the minimum. Taking expectations and using the OLS sampling variance $\mathrm{Var}(\hat{h}) = \sigma_\varepsilon^2/(n\sigma_F^2)$,

$$E\big[v(\hat{h})\big] = \sigma_\varepsilon^2 \left(1 + \frac{1}{n}\right) \tag{TN3.5}$$

and with k contracts estimated jointly the same argument gives a factor $(1 + k/n)$.

Every extra hedging instrument therefore carries a fixed charge of $1/n$ *of the residual variance*, paid whether or not the instrument helps. Comparing it against what the instrument earns settles a question the population formula (TN3.4) cannot: *whether to use it at all*.

Example 4.1 The second contract does not cover its cost

Take the airline's realistic case, $\rho_{12} = 0.90$, and three years of monthly data, $n = 36$.

	population	$\times(1 + k/n)$	**expected**
Heating oil alone ($k = 1$)	3.6864×10^{-4}	1.0278	3.7888×10^{-4}
Heating oil and crude ($k = 2$)	3.6379×10^{-4}	1.0556	3.8400×10^{-4}

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The two-contract hedge is expected to be worse than the one-contract hedge, despite fitting the sample better—as it must, since adding a regressor never raises the residual sum of squares. The 1.3% reduction in population variance does not cover the 2.8% estimation charge.

Two thresholds follow. Holding $\rho_{12} = 0.90$, the second contract pays only once $n > 74$ monthly observations—*more than six years of data*, over which the ratio itself is unlikely to have stayed still. Holding $n = 36$, it pays only if $\rho_{12} < 0.880$.

This is the quantitative form of a warning Section 3.5 gives qualitatively. A hedge ratio estimated over a long sample is precise and may be irrelevant; one estimated over a short sample is relevant and imprecise, and (TN3.5) prices the second half of that trade-off.

5 TAILING THE HEDGE

Equation (3.4) counts contracts in units of the underlying. Futures are settled daily, so the correction is not cosmetic.

Suppose the hedge runs to T and the futures position is N contracts of Q_F units. A mark-to-market gain at time t is received at t and earns interest to T, so its contribution at T is $N Q_F \, \Delta F_t \, e^{r(T-t)}$. The exposure being hedged, by contrast, is settled once, at T. Matching the two requires

$$N Q_F e^{r(T-t)} = h^* Q_A \qquad \Longrightarrow \qquad N^* = \frac{h^* Q_A}{Q_F} \, e^{-r(T-t)}$$

So tailing is a discount factor, and it is the daily settlement of Section 5.12 seen from the hedger's side rather than the arbitrageur's. Its size is easy to misjudge in both directions:

Horizon	1 month	6 months	12 months	24 months
$r = 2\%$	0.9983	0.9900	0.9802	0.9608
$r = 4\%$	0.9967	0.9802	0.9608	0.9231

On Example 3.3's one-month hedge the factor is 0.9967 and 30.5 contracts become 30.4—*below the rounding the hedger must do anyway*. On a two-year hedge at 4% it removes 2.3 contracts from 30.5, which is not negligible. *Tailing is a long-horizon adjustment, and treating it as a universal refinement misplaces the effort.*

6 WHEN VARIANCE IS NOT THE OBJECTIVE

Minimizing variance assumes the hedger is indifferent to the level of the outcome. A hedger who is not solves

$$\max_{h} \; E[\Delta\Pi] - \frac{\lambda}{2} \operatorname{Var}(\Delta\Pi), \qquad E[\Delta\Pi] = \mu_S - h\,\mu_F$$

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where $\mu_F = E[\Delta F]$ and $\lambda > 0$ is risk aversion. Differentiating and using (TN3.2),

$$h^*_\lambda = \underbrace{\rho \, \frac{\sigma_S}{\sigma_F}}_{\text{hedging}} - \underbrace{\frac{\mu_F}{\lambda \, \sigma_F^2}}_{\text{speculative}} \tag{TN3.6}$$

The minimum-variance ratio is the special case in which the second term vanishes, which happens when the hedger is infinitely risk averse or—the case that matters—when the futures price is a martingale, $\mu_F = 0$. Section 5.13 is about whether it is.

Written this way the formula draws the line the chapter's Section 3.7 draws in words. A firm hedging less than h^* is either taking the second term deliberately, which is a view on μ_F and should be defended as one, or is solving a different problem altogether—an uncertain quantity to hedge, a cash constraint, a shareholder who bought the exposure on purpose. *Only the first of those is described by* (TN3.6), and confusing it with the others is how a hedging policy becomes a position no one has approved.

REFERENCES

Johnson (1960) and Stein (1961) are where the portfolio treatment of hedging begins, and Ederington (1979) is the paper that made R^2 the standard measure of hedge effectiveness and applied it to the then-new financial futures. For the estimation issues of Section 4, any careful econometrics text covers the sampling distribution of the OLS slope; the specific point that in-sample fit overstates out-of-sample hedging performance is the regression version of a general result and is worth knowing in that form. Hull's *Options, Futures, and Other Derivatives* states the adjustment in the value form, $N^* = \hat{h} V_A / V_F$ with $\hat{h}$ estimated on percentage changes. That form and the discount factor of Section 5 are not the same correction and should not be conflated: the value form carries no interest rate and is simply what a ratio estimated on returns requires, while the discount factor is the interest earned on variation margin and survives even when rates are certain. Equation (3.6) of the chapter gives the value form and Section 5 derives the other.

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